



Sincronización NTP

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Sincronización NTP

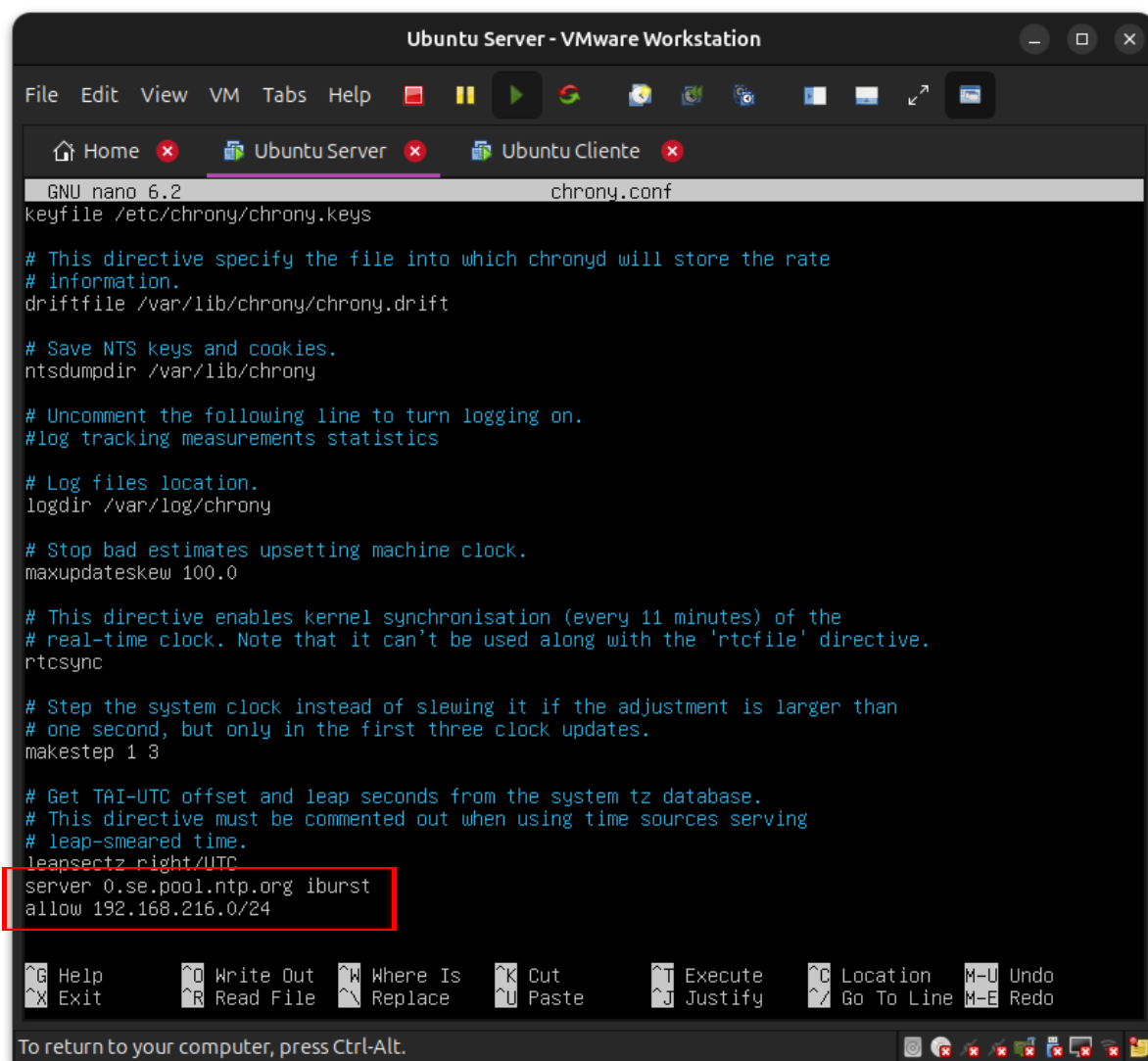
1. Instalación Chrony en Servidor

```
Ubuntu Server - VMware Workstation
File Edit View VM Tabs Help
Home x Ubuntu Server x Ubuntu Cliente x
ivan@ivaninstituto:~$ sudo apt install chrony -y
Leyendo lista de paquetes... Hecho
Creando árbol de dependencias... Hecho
Leyendo la información de estado... Hecho
Los paquetes indicados a continuación se instalaron de forma automática y ya no son necesarios.
  linux-headers-5.15.0-163 linux-headers-5.15.0-163-generic linux-headers-5.15.0-164
  linux-headers-5.15.0-164-generic linux-image-5.15.0-163-generic linux-image-5.15.0-164-generic
  linux-modules-5.15.0-163-generic linux-modules-5.15.0-164-generic
  linux-modules-extra-5.15.0-163-generic linux-modules-extra-5.15.0-164-generic
Utilice «sudo apt autoremove» para eliminarlos.
Los siguientes paquetes se ELIMINARÁN:
  systemd-timesyncd
Se instalarán los siguientes paquetes NUEVOS:
  chrony
0 actualizados, 1 nuevos se instalarán, 1 para eliminar y 0 no actualizados.
Se necesita descargar 290 kB de archivos.
Se utilizarán 360 kB de espacio de disco adicional después de esta operación.
Des:1 http://es.archive.ubuntu.com/ubuntu jammy/main amd64 chrony amd64 4.2-2ubuntu2 [290 kB]
Descargados 290 kB en 2s (189 kB/s)
(Leyendo la base de datos ... 228780 ficheros o directorios instalados actualmente.)
Desinstalando systemd-timesyncd (249.11-0ubuntu3.17) ...

Progreso: [ 29%] [#####.....]

To return to your computer, press Ctrl-Alt.
```

Para instalar el servicio de Chrony tendremos que ejecutar el **comando**
sudo apt install chrony -y



```
GNU nano 6.2 chrony.conf
keyfile /etc/chrony/chrony.keys

# This directive specify the file into which chronyd will store the rate
# information.
driftfile /var/lib/chrony/chrony.drift

# Save NTS keys and cookies.
ntsdumpdir /var/lib/chrony

# Uncomment the following line to turn logging on.
#log tracking measurements statistics

# Log files location.
logdir /var/log/chrony

# Stop bad estimates upsetting machine clock.
maxupdateskew 100.0

# This directive enables kernel synchronisation (every 11 minutes) of the
# real-time clock. Note that it can't be used along with the 'rtcfile' directive.
rtcsync

# Step the system clock instead of slewing it if the adjustment is larger than
# one second, but only in the first three clock updates.
makestep 1 3

# Get TAI-UTC offset and leap seconds from the system tz database.
# This directive must be commented out when using time sources serving
# leap-smearred time.
#leapsectz right/UTC
server 0.se.pool.ntp.org iburst
allow 192.168.216.0/24

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute  ^C Location  M-U Undo
^X Exit      ^R Read File ^N Replace   ^U Paste     ^J Justify  ^_ Go To Line M-E Redo

To return to your computer, press Ctrl-Alt.
```

Cuando lo tengamos instalado tendremos que configurar el archivo de
chrony.conf

Para eso ejecutaremos el **comando** `sudo nano /etc/chrony/chrony.conf`

En este archivo añadiremos las siguientes líneas:

server 0.se.pool.ntp.org **iburst** para añadir un servidor global

allow **192.168.216.0/24** que es el rango de nuestra IP

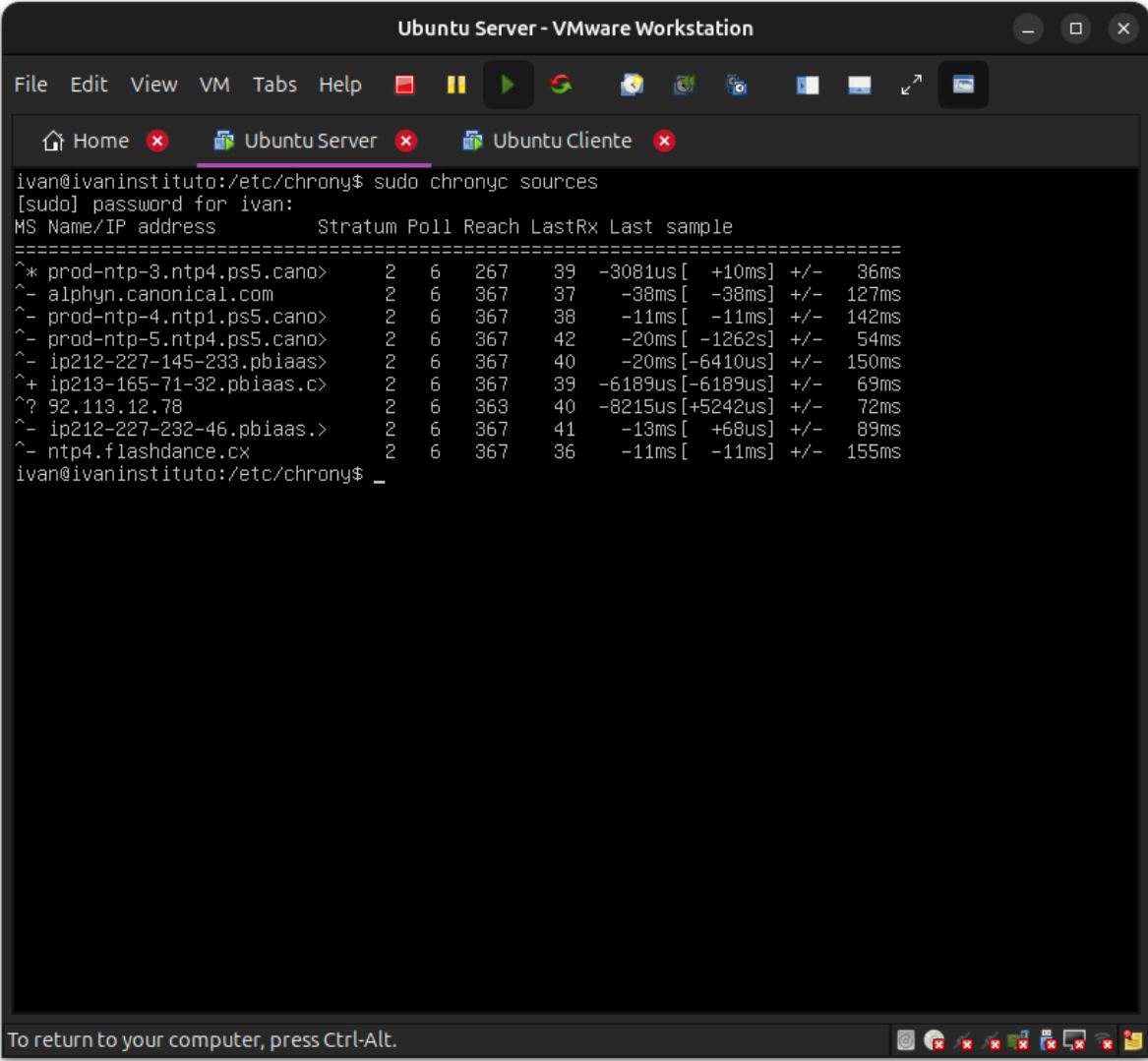


```
Ubuntu Server - VMware Workstation
File Edit View VM Tabs Help
Home x Ubuntu Server x Ubuntu Cliente x
ivan@ivaninstituto:/etc/chrony$ ls
chrony.conf  chrony.keys  conf.d  sources.d
ivan@ivaninstituto:/etc/chrony$ sudo systemctl status cr
cron.service  cryptdisks.service  cryptsetup.target
cryptdisks-early.service  cryptsetup-pre.target
ivan@ivaninstituto:/etc/chrony$ sudo systemctl status chronyd
• chrony.service - chrony, an NTP client/server
   Loaded: loaded (/lib/systemd/system/chrony.service; enabled; vendor preset: enabled)
   Active: active (running) since Wed 2026-02-11 14:32:02 UTC; 1min 6s ago
     Docs: man:chronyd(8)
           man:chronyc(1)
           man:chrony.conf(5)
   Process: 16681 ExecStart=/usr/lib/systemd/scripts/chronyd-starter.sh $DAEMON_OPTS (code=exited,
 Main PID: 16691 (chronyd)
    Tasks: 2 (limit: 4512)
   Memory: 1.4M
      CPU: 93ms
   CGroup: /system.slice/chrony.service
           └─16691 /usr/sbin/chronyd -F 1
             └─16692 /usr/sbin/chronyd -F 1

feb 11 14:32:02 ivaninstituto systemd[1]: Starting chrony, an NTP client/server...
feb 11 14:32:02 ivaninstituto chronyd[16691]: chronyd version 4.2 starting (+CMDMON +NTP +REFCLOCK >
feb 11 14:32:02 ivaninstituto chronyd[16691]: Frequency -7.182 +/- 70.225 ppm read from /var/lib/ch>
feb 11 14:32:02 ivaninstituto chronyd[16691]: Using right/UTC timezone to obtain leap second data
feb 11 14:32:02 ivaninstituto chronyd[16691]: Loaded seccomp filter (level 1)
feb 11 14:32:02 ivaninstituto systemd[1]: Started chrony, an NTP client/server.
feb 11 14:32:09 ivaninstituto chronyd[16691]: Selected source 85.30.141.125 (0.se.pool.ntp.org)
feb 11 14:32:09 ivaninstituto chronyd[16691]: System clock TAI offset set to 37 seconds
feb 11 14:32:10 ivaninstituto chronyd[16691]: Source 185.198.109.227 replaced with 208.85.20.220 (1>
feb 11 14:32:11 ivaninstituto chronyd[16691]: Selected source 185.125.190.58 (ntp.ubuntu.com)
lines 1-25/25 (END)

To return to your computer, press Ctrl-Alt.
```

Ahora reiniciamos el servicio con el **comando** **sudo systemctl restart chronyd** y luego ejecutaremos el **comando** **sudo systemctl status chronyd** y si nos sale **active (running)** en color verde es porque está funcionando correctamente

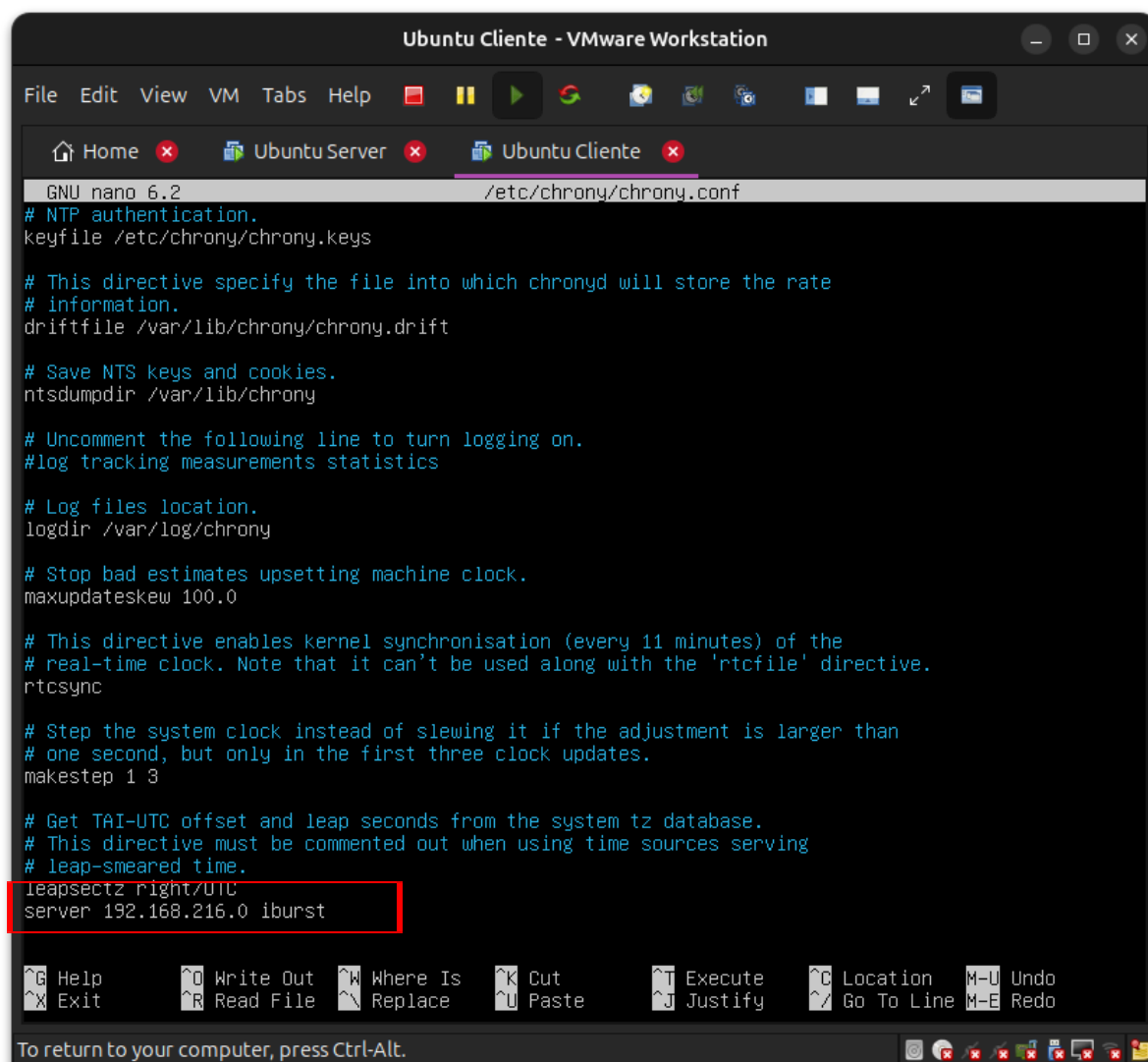


```
ivan@ivaninstituto:/etc/chrony$ sudo chronyc sources
[sudo] password for ivan:
MS Name/IP address         Stratum Poll Reach LastRx Last sample
=====
^* prod-ntp-3.ntp4.ps5.cano> 2 6 267 39 -3081us[ +10ms] +/- 36ms
^- alphyn.canonical.com     2 6 367 37 -38ms[ -38ms] +/- 127ms
^- prod-ntp-4.ntp1.ps5.cano> 2 6 367 38 -11ms[ -11ms] +/- 142ms
^- prod-ntp-5.ntp4.ps5.cano> 2 6 367 42 -20ms[ -1262s] +/- 54ms
^- ip212-227-145-233.pbiaas> 2 6 367 40 -20ms[-6410us] +/- 150ms
^+ ip213-165-71-32.pbiaas.c> 2 6 367 39 -6189us[-6189us] +/- 69ms
^? 92.113.12.78              2 6 363 40 -8215us[+5242us] +/- 72ms
^- ip212-227-232-46.pbiaas.> 2 6 367 41 -13ms[ +68us] +/- 89ms
^- ntp4.flashdance.cx        2 6 367 36 -11ms[ -11ms] +/- 155ms
ivan@ivaninstituto:/etc/chrony$ _
```

Podemos comprobar que el servicio de chronys está sincronizado correctamente con el **comando sudo chronys sources**

2. Instalación Chrony en Cliente

Tendremos que volver a instalar el servicio de chrony pero ahora en nuestra máquina cliente con el **comando sudo apt install chrony -y**



```
GNU nano 6.2 /etc/chrony/chrony.conf
# NTP authentication.
keyfile /etc/chrony/chrony.keys

# This directive specify the file into which chronyd will store the rate
# information.
driftfile /var/lib/chrony/chrony.drift

# Save NTS keys and cookies.
ntsdumpdir /var/lib/chrony

# Uncomment the following line to turn logging on.
#log tracking measurements statistics

# Log files location.
logdir /var/log/chrony

# Stop bad estimates upsetting machine clock.
maxupdateskew 100.0

# This directive enables kernel synchronisation (every 11 minutes) of the
# real-time clock. Note that it can't be used along with the 'rtcfile' directive.
rtcsync

# Step the system clock instead of slewing it if the adjustment is larger than
# one second, but only in the first three clock updates.
makestep 1 3

# Get TAI-UTC offset and leap seconds from the system tz database.
# This directive must be commented out when using time sources serving
# leap-smeared time.
leapsectz right/UTC
server 192.168.216.0 iburst
```

Ahora configuraremos el archivo de configuración de chronys con el **comando `sudo nano /etc/chronys/chrony.conf`**

Añadiremos las siguientes líneas:

server 192.168.216.0 iburst que es la IP de nuestro servidor



```
Ubuntu Cliente - VMware Workstation
File Edit View VM Tabs Help
Home x Ubuntu Server x Ubuntu Cliente x

# Step the system clock instead of slewing it if the adjustment is larger than
# one second, but only in the first three clock updates.
makestep 1 3

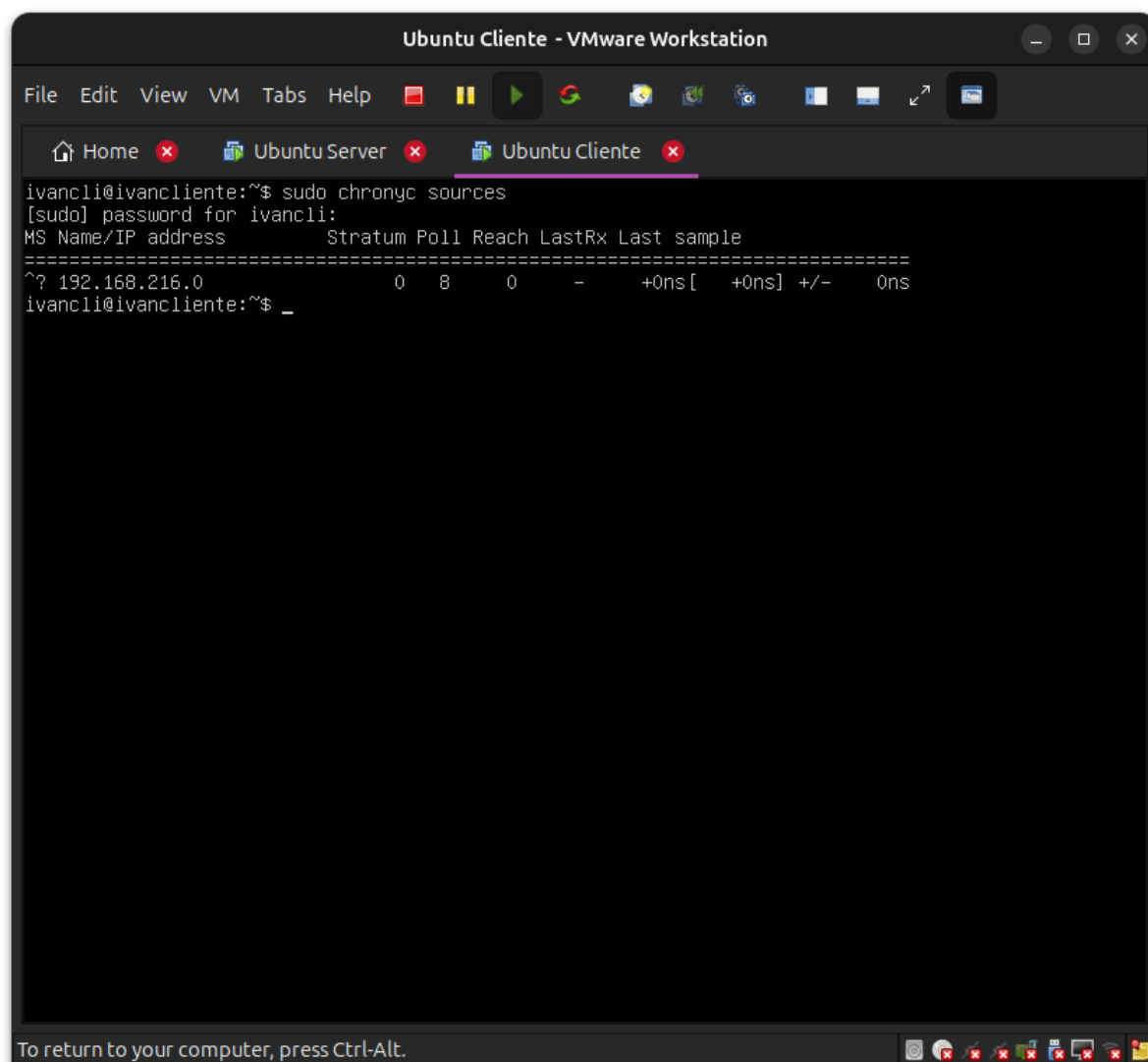
# Get TAI-UTC offset and leap seconds from the system tz database.
# This directive must be commented out when using time sources serving
# leap-smeared time.
leapsectz right/UTC
server 192.168.216.0 iburst

ivancli@ivancliente:~$ sudo systemctl restart chronyd
ivancli@ivancliente:~$ sudo systemctl status chronyd
• chrony.service - chrony, an NTP client/server
   Loaded: loaded (/lib/systemd/system/chrony.service; enabled; vendor preset: enabled)
   Active: active (running) since Wed 2026-02-11 14:38:11 UTC; 7s ago
     Docs: man:chronyd(8)
           man:chronyc(1)
           man:chrony.conf(5)
   Process: 15722 ExecStart=/usr/lib/systemd/scripts/chronyd-starter.sh $DAEMON_OPTS (code=exited,
 Main PID: 15732 (chronyd)
    Tasks: 2 (limit: 4512)
   Memory: 1.3M
      CPU: 51ms
   CGroup: /system.slice/chrony.service
           └─15732 /usr/sbin/chronyd -F 1
             15733 /usr/sbin/chronyd -F 1

feb 11 14:38:11 ivancliente systemd[1]: Starting chrony, an NTP client/server...
feb 11 14:38:11 ivancliente chronyd[15732]: chronyd version 4.2 starting (+CMDMON +NTP +REFCLOCK +R
feb 11 14:38:11 ivancliente chronyd[15732]: Frequency 451.164 +/- 1000000.000 ppm read from /var/lib
feb 11 14:38:11 ivancliente chronyd[15732]: Using right/UTC timezone to obtain leap second data
feb 11 14:38:11 ivancliente chronyd[15732]: Loaded seccomp filter (level 1)
feb 11 14:38:11 ivancliente systemd[1]: Started chrony, an NTP client/server.
lines 1-21/21 (END)

To return to your computer, press Ctrl-Alt.
```

Reiniciaremos el servicio con el **comando `sudo systemctl restart chronyd`** y verificaremos que funcione correctamente con el **comando `sudo systemctl status chronyd`** y si sale **active (running)** en color verde es porque está funcionando correctamente



```
ivanccli@ivanccliente:~$ sudo chronyc sources
[sudo] password for ivanccli:
MS Name/IP address          Stratum Poll Reach LastRx Last sample
=====
^? 192.168.216.0              0      8      0      -      +0ns[  +0ns] +/-    0ns
ivanccli@ivanccliente:~$ _
```

To return to your computer, press Ctrl-Alt.

Por último podemos verificar que estamos conectamos al servidor con el **comando sudo chronyc sources** y si lo hemos configurado todo bien tendremos que ver la IP de nuestro servidor